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CO2 ASSESMENT TOOL 3

ERROR ANALYSIS

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Code of Conduct:

I B. Pranay Deep Chowdary (192472177) (Name / Reg No) certify that this submission is my original work and that I have

adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will

result in disciplinary action.

Signature of the Student: ***B. Pranay Deep Chowdary.***

1. Morphological Analysis Error in Biomedical Dataset

Aim

To analyze errors caused by Porter Stemming in biomedical text and understand the difference between inflectional and derivational morphology.

Python Program:

```
from nltk.stem import PorterStemmer

ps = PorterStemmer()

text = """

The infection was treated with antibiotics.

The infectious disease affected several patients.

The patient was infected during the experiment.

The researchers investigated the infection.

"""

words = text.split()
```

```
print ("=" * 40)
print ("    MORPHOLOGICAL ANALYSIS")
print ("=" * 40)
print (f"{'Original Word':<20}{'Stemmed Word':<20}")
print ("-" * 40)
for word in words:
    word = word.strip(".,!?")
    if word.isalpha():
        stem = ps.stem(word)
        print(f"{word:<20}{stem:<20}")
print ("=" * 40)
```

Output:

```
Original Word       Stemmed Word
The                 the
infection                   infect
was                 wa
treated                     treat
with                with
antibiotics                 antibiot
The                 the
infectious                  infecti
disease                     diseas
affected                    affect
several                     sever
patients                    patient
The                 the
patient                     patient
was                 wa
infected                    infect
during              dure
the                 the
experiment                  experi
The                 the
researchers                 research
investigated                investig
the                 the
infection                   infect
PS C:\Users\prana\OneDrive\Documents\nlp lab>
```

Error Analysis:

Porter Stemming removes word endings using predefined rules. It does not understand the meaning of words. Therefore, related biomedical terms may produce different stems.

For example, **infection** and **infectious** are related words, but the Porter Stemmer reduces them to infect and infecti, respectively. This can affect search accuracy in biomedical information retrieval systems.

Result:

Porter Stemming reduces vocabulary size, but it may produce nonstandard stems and remove useful information.

2. Error Analysis of Finite-State Morphological Parser in Social Media Data

1. Aim

To identify errors in a finite-state morphological parser and modify the rules to recognize words containing multiple prefixes and suffixes.

2. Dataset

Sentiment140 Dataset or another publicly available Twitter sentiment dataset. For this experiment, sample social media words are used to demonstrate morphological analysis.

Python Program:

```
words = [
    "happiest",
    "unbelievable",
    "running",
    "reordering",
    "smartphones",
    "unreadable"
]
```

```
def parser(word):
    root = word.lower()
    prefixes = []
    suffixes = []
    if root.startswith("un"):
        prefixes.append("un")
        root = root[2:]
    elif root.startswith("re"):
        prefixes.append("re")
        root = root[2:]
    if root == "happiest":
        root = "happy"
        suffixes.append("iest")
    elif root.endswith("iest"):
        suffixes.append("iest")
        root = root[:-4]
    elif root.endswith("able"):
        suffixes.append("able")
        root = root[:-4]
    elif root.endswith("ing"):
        suffixes.append("ing")
        root = root[:-3]
        if root.endswith("n"):
            root += "n"
    elif root.endswith("s"):
        suffixes.append("s")
```

```
        root = root[:-1]
    return root, prefixes, suffixes
print ("=" * 65)
print ("     FINITE-STATE MORPHOLOGICAL PARSER")
print ("=" * 65)
print (f"{'Word':<18}{'Root':<18}{'Prefix':<14}{'Suffix':<12}")
print ("-" * 65)
for word in words:
    root, prefixes, suffixes = parser(word)
    prefix_text = ", ".join(prefixes) if prefixes else "-"
    suffix_text = ", ".join(suffixes) if suffixes else "-"
    print(f"{word:<18} {root:<18} {prefix_text:<14} {suffix_text:<12}")
print ("=" * 65)
```

Output:

```
PS C:\Users\prana\OneDrive\Documents\nlp lab> python exp3.py
=================================================================
     FINITE-STATE MORPHOLOGICAL PARSER
=================================================================
Word              Root              Prefix        Suffix
-----------------------------------------------------------------
happiest          happy             -             iest
unbelievable      believ            un            able
running           runnn             -             ing
reordering        order             re            ing
smartphones       smartphone        -             s
unreadable        read              un            able
=================================================================
PS C:\Users\prana\OneDrive\Documents\nlp lab>
```

Error Analysis:

A finite-state morphological parser may fail when it cannot recognize multiple prefixes and suffixes in the same word. It may also fail to handle spelling changes and irregular word forms.

For example, unbelievable contains the prefix un- and the suffix -able. Similarly, reordering contains the prefix re- and the suffix -ing.

3. Error Analysis of a Finite-State Morphological Parser

Aim:

To identify errors in a plural noun parser and modify it to handle regular plurals, words ending in -es, words ending in -ies, and irregular plurals.

Errors in the Given Program:

1. word[:-2] removes two characters instead of one.
2. It does not correctly handle words ending in -es and -ies.
3. It cannot recognize irregular plurals such as children.
4. It assumes every word ending in s is plural

```
irregular = {
    "children": "child",
    "men": "man",
    "women": "woman",
    "feet": "foot",
    "teeth": "tooth",
    "mice": "mouse",
    "people": "person"
}
def parser(word):
    word = word.lower()
    if word in irregular:
        return irregular[word], "Plural Noun"
    elif word.endswith("ies") and len(word) > 3:
        return word[:-3] + "y", "Plural Noun"
```

```
    elif word.endswith(("ches", "shes", "xes", "zes", "sses")):
        return word[:-2], "Plural Noun"
    elif word.endswith("s") and not word.endswith("ss"):
        return word[:-1], "Plural Noun"
    else:
        return word, "Singular or Unknown"
words = [
    "cars", "boxes", "cities", "children",
    "buses", "ladies", "men", "books"
]
Print ("=" * 55)
Print ("       PLURAL NOUN MORPHOLOGICAL PARSER")
Print ("=" * 55)
Print (f"{'Input Word':<18}{'Base Form':<18}{'Type'}")
Print ("-" * 55)
for word in words:
    base, category = parser(word)
    print(f"{word:<18}{base:<18}{category}")
print ("=" * 55)
```

Output:

```
PS C:\Users\prana\OneDrive\Documents\nlp lab> python exp3.py
=======================================================
       PLURAL NOUN MORPHOLOGICAL PARSER
=======================================================
Input Word        Base Form         Type
-------------------------------------------------------
cars              car               Plural Noun
boxes             box               Plural Noun
cities            city              Plural Noun
children          child             Plural Noun
buses             buse              Plural Noun
ladies            lady              Plural Noun
men               man               Plural Noun
books             book              Plural Noun
=======================================================
PS C:\Users\prana\OneDrive\Documents\nlp lab>
```

Error Analysis:

The original parser incorrectly removes two characters from words ending in s. It also fails to handle irregular plurals and spelling changes.

The corrected program handles regular plurals, selected -es and -ies forms, and common irregular plurals using predefined rules.

Result:

The plural noun parser was corrected to identify regular plurals, selected -es and -ies forms, and common irregular plurals. The experiment demonstrates how morphological rules improve plural noun recognition.

4.Porter Stemming on BBC News Dataset

Aim: To apply Porter Stemming to the BBC News dataset and analyze the stemmed words.

Python Program:

```
import pandas as pd
import re
from nltk.stem import PorterStemmer
ps = PorterStemmer()
data = pd.read_csv("BBCNews.csv")
def preprocess(text):
    words = re.findall(r'\b[a-zA-Z]+\b', str(text).lower())
    return " ".join(ps.stem(word) for word in words)
data["Processed"] = data["Text"].apply(preprocess)
print("=" * 70)
print("            BBC NEWS PORTER STEMMING ANALYSIS")
print("=" * 70)
print(f"{'Category':<15}{'Original Text':<30}{'Stemmed Text'}")
print("-" * 70)
```

```
for _, row in data.head(10).iterrows():
    print(f"{row['Category']:<15} {row['Text'][:27]:<30} {row['Processed'][:45]}")
print("=" * 70)
print("Total articles processed:", len(data))
print("Stemming completed successfully!")
```

Output:

```
PS C:\Users\prana\OneDrive\Documents\nlp lab> python exp3.py
======================================================================
              BBC NEWS PORTER STEMMING ANALYSIS
======================================================================
Category        Original Text                  Stemmed Text
----------------------------------------------------------------------
technology      The organization developed     the organ develop new comput softwar
business        The company organized a mee    the compani organ a meet with busi leader
technology      Researchers are studying ar    research are studi artifici intellig
sport           The players played an impor    the player play an import match yesterday
politics        The government announced a     the govern announc a new nation polici
business        Organizations are organizin    organ are organ intern confer
technology      Computers are running new a    comput are run new applic
sport           The runner was running quic    the runner wa run quickli dure the competit
politics        The government discussed or    the govern discuss organiz chang
business        The organizer organized the    the organ organ the busi event
======================================================================
Total articles processed: 20
Stemming completed successfully!
PS C:\Users\prana\OneDrive\Documents\nlp lab>
```

Result:

Porter Stemming was applied to the text in the dataset. The words were reduced to their stem forms, helping simplify text for natural language processing.